CAM using SolidWorks

Sessions	Topic	Type	Duration (40 Hours)
Session 1	**Introduction to CAM & SolidWorks CAM**	Experiential learning	2.0
	Introduction to CAM		
	Overview of SolidWorks CAM		
	Benefits of Solidworks CAM		
	Manufacturing Process		
	Integration with CAD (Design-to-Manufacturing workflow)		
	CNC basics (Machines, axes, coordinates)		
	Milling machines overview		
	Turning machine overview		
	Closed-loop manufacturing concept		
	Solid Modeling Tools		
	Feature Creation Tools (Base Features)		
	Material Removal Tools (Cut Features)		
	Modification & Detailing Tools		
	Advanced Feature Tools		
	Hole & Pattern Tools		
	Curve & Reference Tools		
Session 2	**Hands On 1:** Creating the Gear Box Case	Practical	2.0
Session 3	**CAM Setup Fundamentals**	Experiential learning	2.0
	Navigating CAM tab & command manager		
	Define Machine Operation (Mill Machine, Mill 4 Axis, Mill 5 Axis,		
	Coordinate systems (WCS, MCS)		
	Stock definition (bounding box, Pre Defined bounding box,		
	Mill Setup		
	Machine Definition (Post + Axis limits + travel)		
	Fixture / Clamp / Vice definition		
	Setup Sheets generation		
	Tool number vs tool ID concept		
	Operation Planning		
	Generate machining sequence using Generate Operation Plan		
	Assign tools using Tool Crib		
	Organize operations in Operation Tree Manager		
Session 4	**Hands On 2:** Car Door Control Panel Trim Core – CAM Setup & WCS	Practical	2.0

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Session 5	**Feature Recognition (Core Module)**	Experiential learning	2.0
	Automatic Feature Recognition		
	Apply Extract Machinable Features (AFR)		
	Identify pockets, holes, slots, bosses automatically		
	Understand feature hierarchy in Feature Tree		
	Edit features using Edit Definition		
	Maintain CAD-CAM associativity		
	Interactive Feature Recognition		
	Create features manually using IFR (Feature Creation Wizard)		
	Select geometry using face/edge filters & chain selection		
	Modify and correct missing features		
	Handle complex geometries not recognized by AFR		
Session 6	**Hands On 3**: Automatic Features Recognition & Editing in Mold	Practical	2.0
	Hands On 4: Interactive Feature Recognition in Dual-Rod Cylinder		
Session 7	**Toolpath Generation**	Experiential learning	2.0
	Generate toolpaths using Generate Toolpath		
	Configure operations in Operation Parameters Panel		
	Apply pocketing (zig-zag / offset), contouring, drilling cycles		
	Control machining using step-over, step-down, feed rate		
	Visualize paths using Toolpath Preview		
	Entry/Exit strategies (lead-in, lead-out)		
	Cut patterns (climb vs conventional)		
	Rest machining concept		
	Tool engagement & chip load control		
	Simulation & Verification		
	Run simulation using Simulate Toolpath		
	Visualize material removal process		
	Detect collisions and errors		
	Validate output using Compare Tool (stock vs model)		
	Compare design vs machined part		
Session 8	**Hands On 5:** Toolpath Simulation and Verification of a Mechanical	Practical	2.0

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Session 9	**Operation Management**	Experiential learning	2.0
	Modify machining sequence using Reorder Operations		
	Organize using Sort Manager (tool / feature based)		
	Reduce tool changes and idle movement		
	Optimize machining cycle time		
	Post Processing & Output		
	Generate CNC program using Post Process Tool		
	Review code using NC Code Viewer		
	Understand G-code (G00, G01, G02, G03) and M-codes		
	Validate machine compatibility and output		
	Toolpath linking optimization		
	Safe clearance plane strategy		
	Rapid move control (G00 safety)		
	Final CAM workflow validation		
Session 10	**Hands On 6:** Hinge Part Manufacturing Operation Management	Practical	2.0
Session 11	**Milling Operation (2.5 Axis)**	Experiential learning	2.0
	Introduction to 2.5 Axis Milling		
	Feature Definition & Feature Add		
	Tool Selection & Tool Library Management		
	Rough Milling Operation (Material Removal Strategy)		
	Contour Milling Operation (Profile Finishing)		
	Face Milling Operation (Surface Preparation)		
	Thread Mill Operation		
	Operation Parameters & Optimization		
	Simulation & Verification and Post Processing & CNC Output		
Session 12	**Hands-On 7:** 2.5 Axis Mill Operation on a Piston Head	Practical	2.0

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Session 13	**Milling Operation (3 Axis)**	Experiential learning	2.0
	Introduction to 3 Axis Milling		
	Feature Definition & Feature Add		
	Area Clearance (3 Axis Roughing Strategies)		
	Z Level (Contour-Based Finishing)		
	Flat Area Machining		
	Operation Parameters & Optimization		
	Simulation & Verification and Post Processing & CNC Output		
	Probe Operation		
	Introduction to Probing in CNC Machining		
	Probe calibration		
	Workpiece alignment (X/Y/Z probing)		
	Tool length measurement		
	Broken tool detection		
	In-process inspection		
	Datum setting using probe cycles		
Session 14	**Hands-On 8:** Hole Machining in Multi Hole Flange	Practical	2.0
Session 15	**Turn Machining**	Experiential learning	2.0
	Introduction to Turn Machining		
	Setup the Turning machine		
	Turning Operation - Face Rough, Face Finsih, Turn Rough, Turn		
	Turning Groove Operation - Groove Rough and Groove Finish		
	Turning Bore Operation - Bore Rough, Bore Finsih, Drill, Center		
	Operation Parameters & Optimization		
	Simulation & Verification and Post Processing & CNC Output		
Session 16	**Hands On 9:** 3 Axis Mill Operation in Cavity Model	Practical	2.0

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Sessions	**Topic**	**Type**	**Duration (40 Hours)**
Session 17	**Hole Machining Operation** Introduction to Hole Machining Hole Feature Recognition & Feature Setup Tool Selection for Hole Machining Drilling Operations (Core Operation) - Drilling, Pecking and Gun Countersink Operation - Spot Drilling and Pecking Bore Operation - Boring and Bore with Dwell Ream Operation - Reaming and Ream W/Dwell Tap Operation - Tapping and Reverse Tapping Rough Milling Operation - Cavity and Core machining Contour Milling Operation - Contour and Contour & Floor Thread Mill Operation Operation Parameters & Optimization Simulation & Verification and Post Processing & CNC Output	Experiential learning	2.0
Session 18	**Hands On 10:** Turning Operation in Turning Machine	Practical	2.0
Session 19	**Advanced Toolpath Control & Optimization** Technology Database (TechDB) – Advanced Usage Tolerance-Based Machining (TBM) Multi-Setup & Indexing Strategies Tool Management – Advanced Concepts Post Processor Customization & Debugging CAM Data Management & Revision Control CNC Code Optimization & Shopfloor Readiness	Experiential learning	2.0
Session 20	**Hands On 11:** Creating Custom Mill Tools and Holders in	Practical	2.0